Pranav Nuti

The University of Chicago Booth School of Business
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Employment

The University of Chicago Booth School of Business **2024–**
Principal Researcher
Postdoctoral-level position, mentored by Rad Niazadeh

Education

Stanford University **2018–2024**
PhD in Mathematics
Advised by Jan Vondrák

Indian Institute of Science (IISc), Bangalore **2014–2018**
Bachelor of Science (Research), Gold Medalist in Mathematics

Research

Journal Publications and Revisions

Stationary Online Contention Resolution Schemes **Operations Research, Major Revision**
with Mohammad Reza Aminian and Rad Niazadeh

Prophet Inequalities with Cancellation Costs **Operations Research, Minor Revision**
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Toward an Optimal Contention Resolution Scheme for Matchings **Mathematical Programming**
with Jan Vondrák

Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities **Operations Research, Minor Revision**
with Will Ma and Calum MacRury

The Secretary Problem with Distributions **Mathematics of Operations Research, Major Revision**

Peer-Reviewed Conference Publications

Stationary Online Contention Resolution Schemes **EC 2026**
with Mohammad Reza Aminian and Rad Niazadeh

Optimal Prophet Inequalities with Buyback **EC 2026**
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Static Pricing for Single Sample Multi-unit Prophet Inequalities **SOSA 2026**
with Peter Westbrook

Online Matching and Contention Resolution for Edge Arrivals with Vanishing Probabilities **EC 2024**
with Will Ma and Calum MacRury

Prophet Inequalities with Cancellation Costs **STOC 2024**
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Secretary Problems: The Power of a Single Sample **SODA 2023**
with Jan Vondrák

A Tight Competitive Ratio for Online Submodular Welfare Maximization **ESA 2023**
with Amit Ganz and Roy Schwartz

Toward an Optimal Contention Resolution Scheme for Matchings **IPCO 2023**
with Jan Vondrák

The Secretary Problem with Distributions **IPCO 2022**

Working Papers

Dynamic Matching with Abandonment: Optimal Guarantees via Static Priority Policies
with Nick Arnosti, Rad Niazadeh, and Felipe Simon

Prophet Inequalities and Online Contention Resolution for Matchoids **(submitted)**
with Calum MacRury and Jan Vondrák

Single-Sample Prophet Inequalities for Knapsack Problems **(submitted)**
with Rebecca Reiffenhäuser and Alice Sayutina

Optimal Bayesian Online Allocation of Reusable Resources
with Mohammad Reza Aminian and Rad Niazadeh

Workshop Papers

Solving Open Problems in Operations Research Using AI **AI-Driven Research in EconCS at EC 2026**
with Eric Fithian and Rad Niazadeh
Draft paper. Based on Open Problems in Operations Research. Also presented as a poster at Can AI do Theory? at STOC 2026.

Bounded Best Buyers: Prophet Inequalities Revisited **Easy Peasy at EC 2026**
with Farbod Ekbatani, Rad Niazadeh, and Jan Vondrák

Magicians Don't Move: An Easy Peasy OCRS **Easy Peasy at EC 2026**
with Mohammad Reza Aminian and Rad Niazadeh

A Match Made by Entropy **Easy Peasy at EC 2026**
with Mohammad Reza Aminian and Rad Niazadeh

Selected Talks

How to Write and Review Papers **2026**
EC 2026 Mentoring Workshop

Prophet Inequalities with Cancellation Costs **2025**
Chicago Junior Theorists Workshop

What is a Good Model for Online Decision Making? **2023**
SURIM Talk, Stanford University

Online Selection Problems and How Ramsey's Theorem Solves a Card Game **2023**
DACO Seminar, ETH Zürich

Between the Secretary Problem and the Prophet Inequality **2022**
Data-Driven Decision Processes Talk, Simons Institute, UC Berkeley

Teaching

Papers

Engaging Activities for Teaching Linear Algebra **PRIMUS 2024**
with Shintaro Fushida-Hardy and Megan Selbach-Allen

Awards

Pólya Teaching Fellow Award **2024**
Awarded for outstanding teaching during the PhD program

Robert Osserman Teaching Award **2022**
Awarded for outstanding contributions as a teaching assistant at Stanford University

Hoefer Mentoring Award **2022**
Awarded for supporting undergraduate writing at Stanford University

Courses Taught as Instructor

Linear Algebra, Stanford Summer Engineering Academy **2021, 2022**
Equity and Inclusion Initiative, Stanford Engineering
This is a summer program that seeks to illuminate the brilliance of students who have been systemically marginalized in engineering. Along with teaching the course, in collaboration with my co-instructors, I also designed the curriculum. 38 classroom hours. Syllabus, lesson plans, handouts and descriptions of all activities are available.

Teaching Outside the University

Stanford Jail and Prison Education Project **2023**
San Francisco County Jail # 5
The project provides education services to people who are incarcerated in Bay Area jails.

International Math Olympiad Training Camp **2015, 2017**
Homi Bhabha Center for Science Education, Mumbai
I taught problem solving, the probabilistic method, elementary number theory, and basic analytic methods in number theory. I also graded exams.

Undergraduate Student Mentorship

Stanford Undergraduate Research in Mathematics **2023**
Department of Mathematics, Stanford University
The program is aimed at introducing Stanford undergraduate students to research in mathematics. I mentored Ethan Zhang, Ezra Steinberg, Peter Westbrook, and Quinn McIntyre. We proved several new results related to prophet inequalities. A report is available.

Directed Reading Program **2020–2022, 2024**
Department of Mathematics, Stanford University
The program is aimed at helping undergraduates, especially members of historically underrepresented groups, participate in the larger mathematical community. I mentored several students: Alessandra Maranca, Sophia Sanchez, Ellen Xu, Donald Poindexter, Logan Bhamidipaty, and Sohan Vichare.

Graduate Student Mentorship

Stanford Summer Engineering Academy **2023**
Equity and Inclusion Initiative, Stanford Engineering
I interviewed instructors to hire for the program, helped design a new curriculum, and provided feedback to instructors.

TA Mentoring Program **2022–2023**
Department of Mathematics, Stanford University
The program is aimed at providing first-time TAs advice and feedback on leading section. I mentored Yosheb Getachew, Joao Campos Vargas, Zhenyuan Zhang, and Hongjian Yang.

Other Awards

Bronze Medal, 54th International Mathematical Olympiad **2013**

KVPY Fellowship **2013–2018**
Awarded by the Govt. of India
A 100% scholarship for undergraduate education.

National Talent Search (NTS) Scholarship **2010–2011**
Awarded by the Govt. of India to high school students